



Computer Architecture

----A Quantitative Approach

2026 Spring



Instructor & TA

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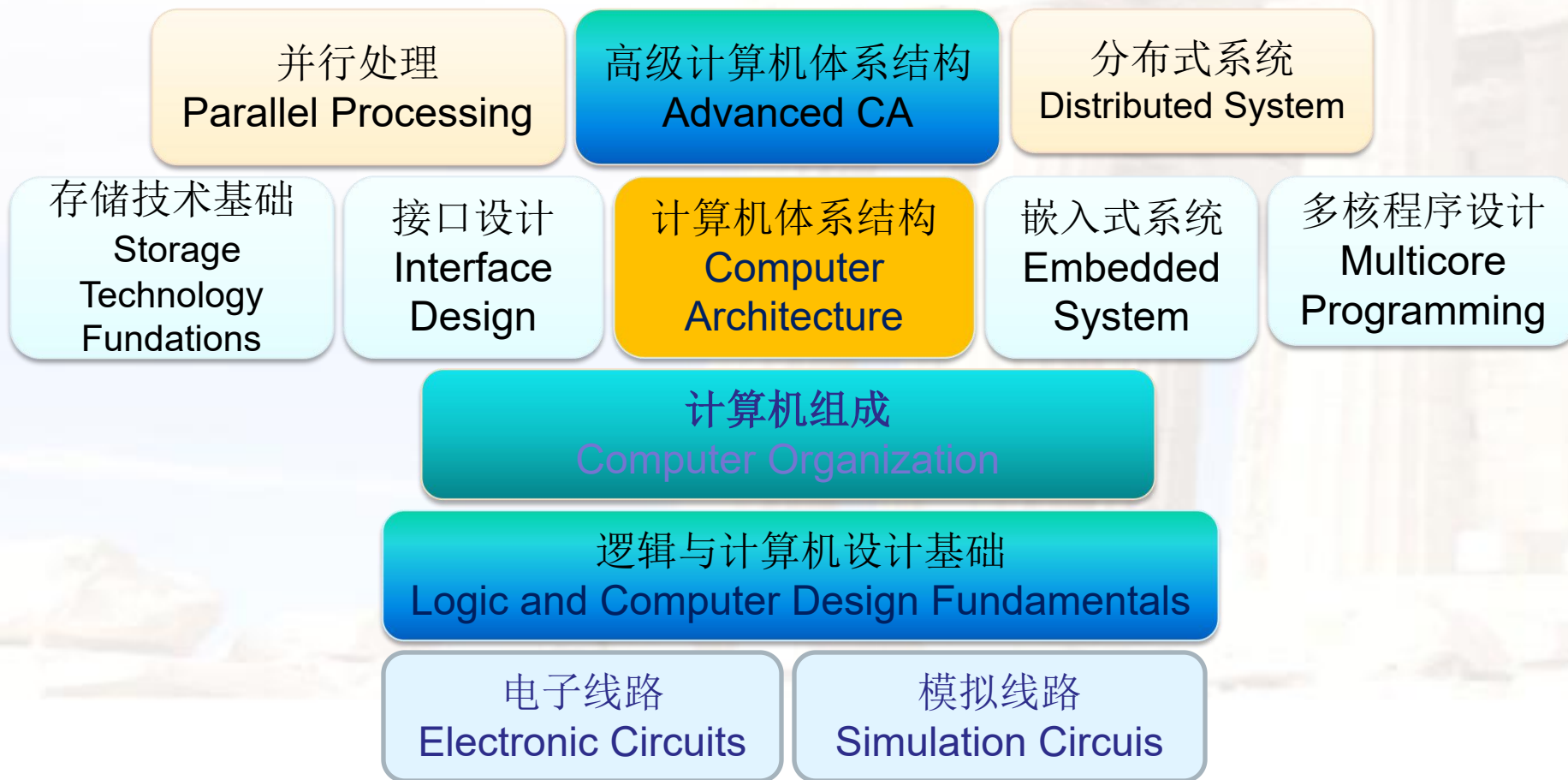
□ Course Website: course.zju.edu.cn





Why we learn Computer Organization?

■ Hardware Course Series





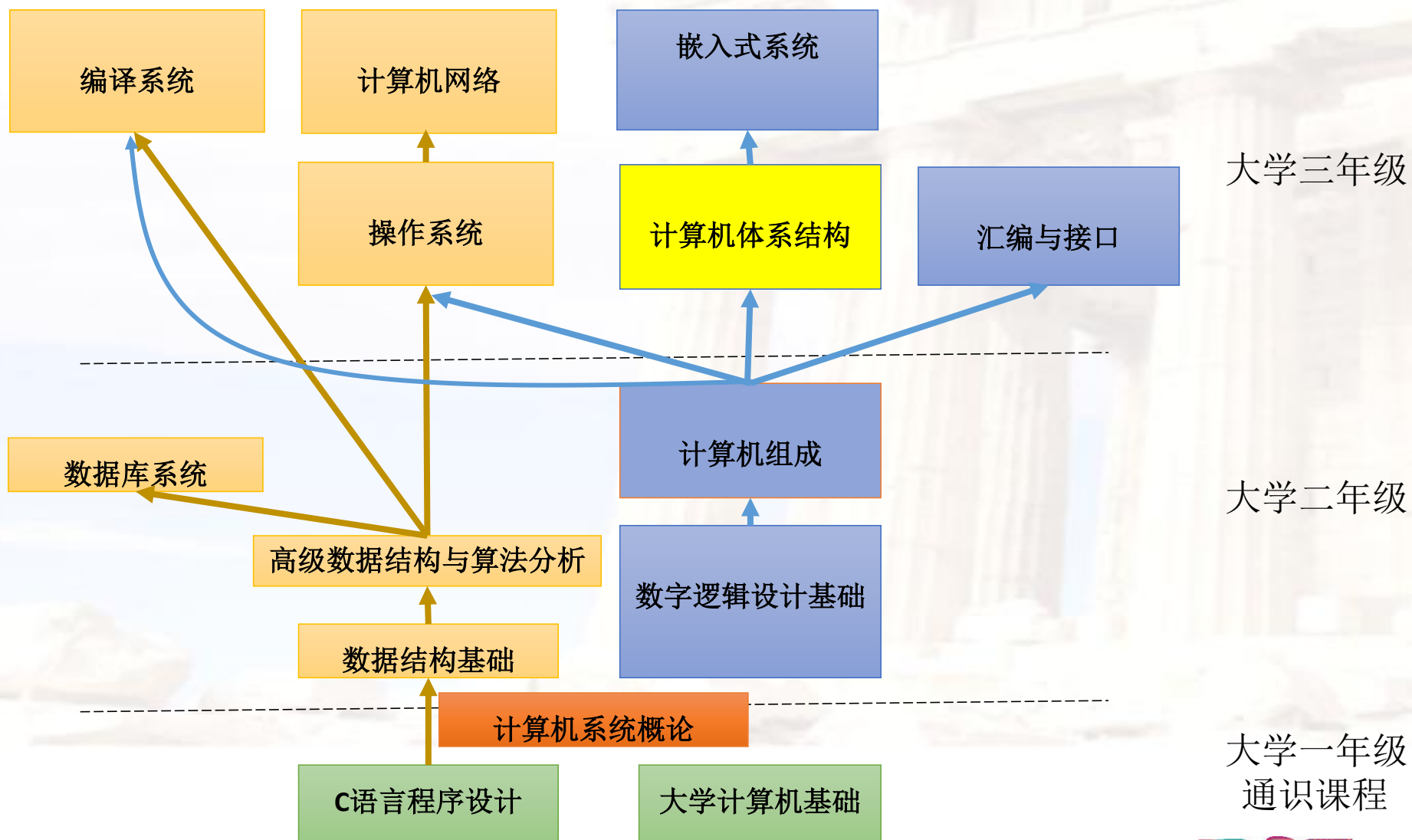
Course Objectives

□ The objective of this course

- systemically learn the fundamental concepts and design approaches of computer architecture from the view of the whole computer system.
- Learn the hardware design approaches and to use hardware design toolkits skillfully.
- learn to implement pipelined CPU and a simple memory hierarchy.



Why we learn Computer Organization?





Textbook

David A. Patterson, John L. Hennessy,

《Computer Architecture – A Quantitative Approach》

6th Edition. July , 2019.





Difference of 6th & 5th edition

❑ Most significant change:

➤ ISA selection

- 5th MIPS
- 6th RISC-V

➤ Add a new chapter:

- Chapter 7 Domain-Specific Architecture(DSA)
- CPU/GPU/DNN accelerator comparison

❑ Other change:

■ Remove description of Delayed Branch

- RISC-V removes it because the performance of branch predictor is powerful enough.

❑ Book size limitation:

- More contents move to Appendix and CD





John L. Hennessy (Stanford)



<https://engineering.stanford.edu/people/john-hennessy>

- Former President of Stanford University during 2000 – 2016 (17 billion)
- Current Alphabet Chairman
- "Godfather of Silicon Valley,"
- In 1981, Hennessy initiated a project at Stanford that focused on a simpler computer architecture known as RISC. During a sabbatical leave in 1984-85 he cofounded MIPS Computer Systems, now known as MIPS Technologies, which specializes in the production of microprocessors SPARC.
- Received [Eckert-Mauchly Award](#) in 2001
- Received [Turing Award](#) in 2017



浙江大學
ZHEJIANG UNIVERSITY



David A. Patterson (UC Berkeley)

- ❑ UC Berkeley (1976 – 2016)
- ❑ Currently Google TPU
- ❑ He led the design and implementation of **RISC** I (the foundation of the SPARC architecture)
- ❑ Inventor of **RAID**
- ❑ involved in the Network of Workstations (NOW) project
- ❑ Research Accelerator for Multiple Processors (RAMP)
- ❑ Received ACM Eckert-Mauchly Award <https://people.eecs.berkeley.edu/~pattsrn/> in ISCA 2008
- ❑ Received Turing Award in 2017





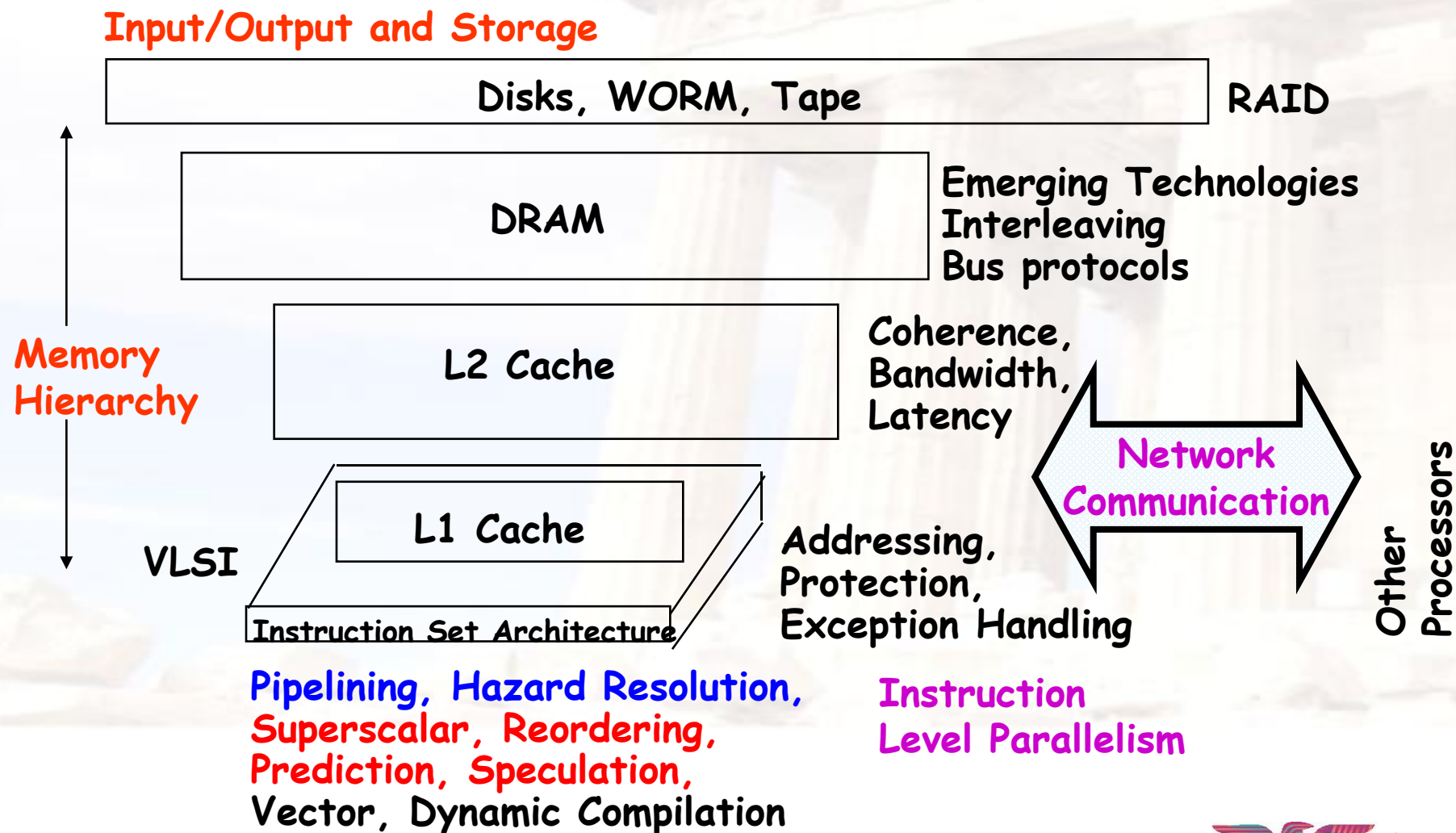
❑ Computer Organization

❑ Survey:

- Digital Logical Design
- Computer Organization
- Operating Systems
- Techniques of Compiling / Fundamentals of Compiling

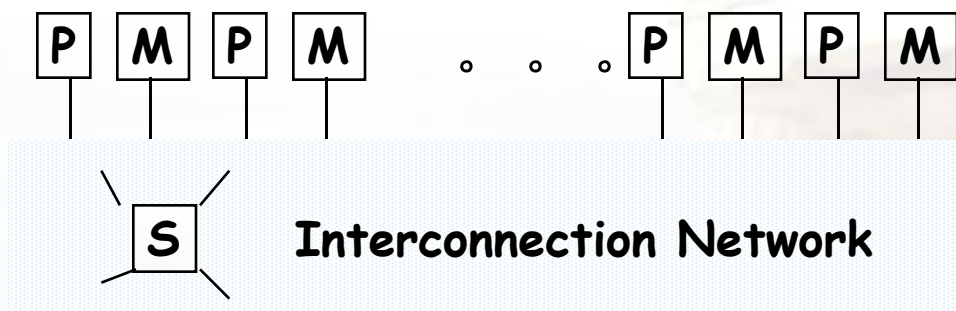


Topics in CA:





Computer Architecture Topics



Processor-Memory-Switch

Multiprocessors
Networks and Interconnections

MCP: Multi-Core Processor
(Chip level Processor)

Shared Memory,
Message Passing,
Data Parallelism

Network Interfaces

Topologies,
Routing,
Bandwidth,
Latency,
Reliability



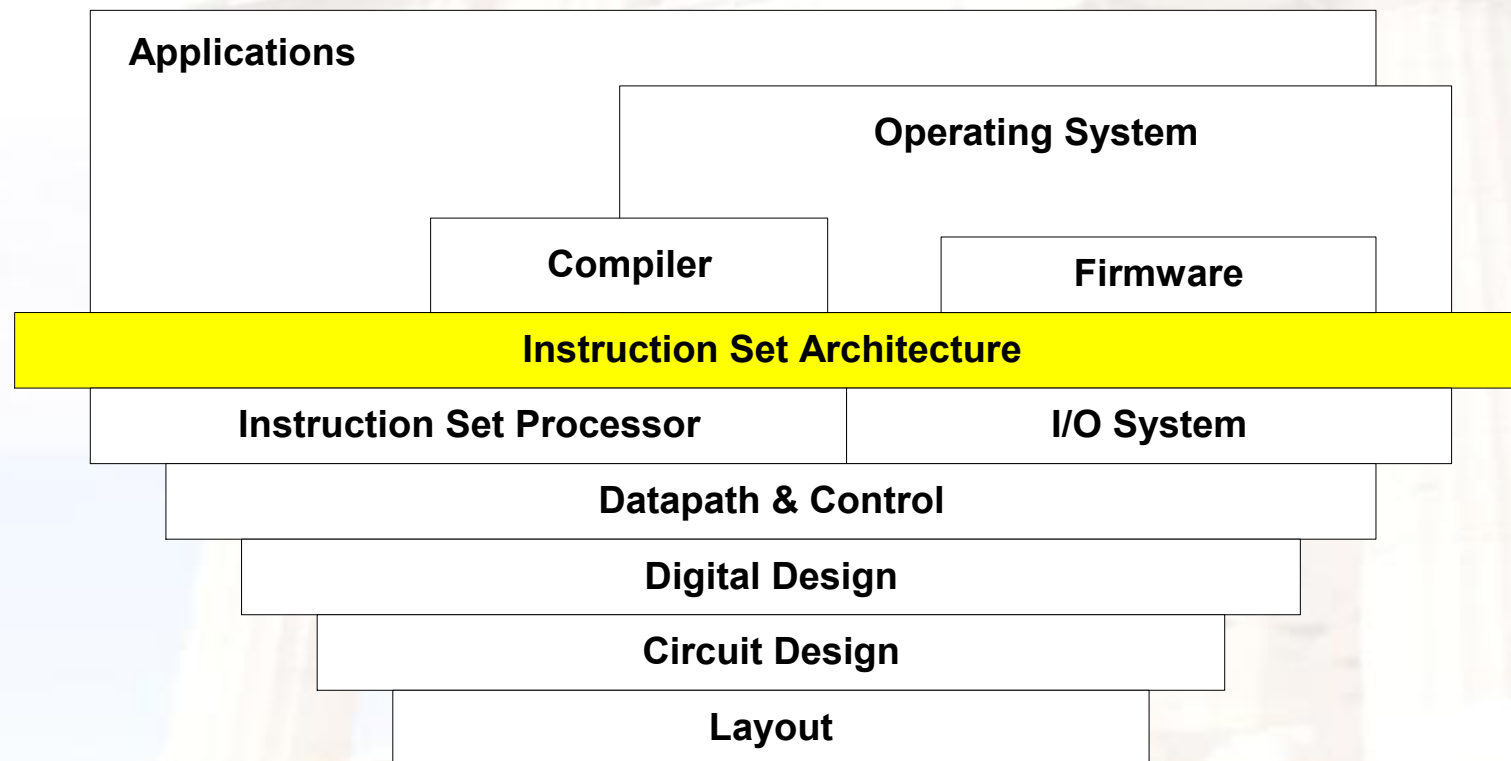
Topics in this Class (6th. Edition)

	Topic	Reference
Ch1.	Fundamentals of Quantitative Design and Analysis (Quantitative principles)	Chapter 1
Review	Pipelining: Basic and Intermediate Concepts (How to design a pipelined CPU)	Appendix C
Ch2.	Memory Hierarchy Design (How to improve cache and memory)	Chapter 2 Appendix B
Ch3.	Instruction-Level Parallelism and Its Exploitation (How to implement dynamic scheduling)	Chapter 3 Appendix A
Ch4.	Data-Level Parallelism in Vector, SIMD, and GPU Architectures	Chapter 4
Ch5.	Thread-Level Parallelism	Chapter 5



Different from Organization

- Part & **Whole**
- Bottom up & **Top down**
- How & **Why**





Course Objectives

□ The objective of this course

- Introduce the fundamental techniques on which high-performance computing is based,
- Develop the foundations for analyzing the benefits of design options in computer architecture.



How ?

- ❑ Concepts, Ideas and Principles
- ❑ Quantitative approaches
- ❑ Hit the problem and right way to solve problem
- ❑ ***As a man sows, so he shall reap.***
一分耕耘一分收获





What we most care about ?

- ❑ For the designer, need to know **why** ?
 - Know the principle & the way to evaluate
 - make decision / make choice.

- ❑ For the designer, need to know universal approach
 - Investigate → problem → solve, make decision → Implement
→ Evaluate



Grading Policy:

- 20%: written homework
 - 40%: Lab assignments
 - 8%: Bonus
 - lab grade = report (40%) + check(60%)
 - 40%: Final exam
 - (close-book test with one A4 memo)
 - Attendance : *deduct one point for each absence*
 - Final grade = 40%(Final exam) + 60%(Lab + Homework) - Attendance deduction.
- } 60% (≤60)



Homeworks (20%)

- ❑ Total 4 times, once per chapter
- ❑ Submission deadline will be normally one week after assigned, and will be announced on **course website**.
- ❑ For doing homework, discussion is greatly encouraged, but every student is required to **Do and Submit** the homework **individually** on time.
- ❑ Submission Naming rule
 - **StID_name_hw1.doc**



Lab assignments 40%

❑ Objective

- Implement a **pipelined CPU** with simplified **interruption** and **memory hierarchy** composed of a memory and a cache supporting **fundamental RISC-V-32I instructions** via Verilog in Xilinx Vivado on SWORD-V4 board

❑ How

- Do the lab **by group with 2 persons** and submit **lab report to website individually**.
- Do the 6 lab assignments **gradually**, examine results each time
- Submit **6 lab reports**

❑ Grading

- Lab1-6: 8%, 6%, 4%, 6%, 8%, 8%



Lab assignments 40%

- **Lab1 (8' 3weeks):** Implement pipelined CPU with forwarding paths and prediction-not-taken supporting RISC V 32i instructions.
- **Lab2 (6' 2weeks):** Implement Interruption and Exception on Pipelined CPU of Lab 1).
- **Lab3 (4' 1week):** Implement a 2-way associative cache
- **Lab4 (6' 2weeks):** Adding the cache into the pipelined CPU in Lab 2).
- **Lab5 (8' 3weeks):** Expend the pipelined CPU to support multi-cycle operations: integer multiplier, integer divider / remainder operation, issue in order and complete out of order, detecting pipeline hazards (WAW, RAW)
- **Lab6 (8' 5weeks):** Implement the Dynamic Scheduling (Scoreboard or Tomasulo), implement a dynamic scheduling pipelined CPU.
- **Bonus (8'):** For novel design and implementation. 2' for each lab and 8' at most.



Submission Policy:

- All the homeworks and lab assignments are required to be submitted to the course website on time.
- **Submission deadline** will be announced on course website.
- All assignments in this course should be turned in by the specified due date. Late assignment will be **penalized 10% every three days late**. However, late assignment **more than 6 days** is **NOT** accepted.



Honest Policy

- Be **HONEST** in your work!
- Found **copy** & **be copied** in the homework or lab report , you get **ZERO** for one submission and also get **10% off** in the final **grade** !



Q&A